

Problem Set 74

Derivatives of Inverse Functions

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Problem 2

Let $f(x) = x \arctan x$. Use the limit definition of derivative to find $f'(\sqrt{3})$.

$$\begin{aligned}
 f'(\sqrt{3}) &= \lim_{h \rightarrow 0} \frac{(\sqrt{3} + h) \arctan(\sqrt{3} + h) - \sqrt{3} \arctan \sqrt{3}}{h} \\
 &= \sqrt{3} \lim_{h \rightarrow 0} \frac{\arctan(\sqrt{3} + h) - \arctan \sqrt{3}}{h} + \lim_{h \rightarrow 0} \arctan(\sqrt{3} + h) \\
 &= \sqrt{3} \lim_{h \rightarrow 0} \frac{\arctan\left(\frac{\sqrt{3} + h - \sqrt{3}}{1 + \sqrt{3}(\sqrt{3} + h)}\right)}{h} + \frac{\pi}{3} \\
 &= \sqrt{3} \lim_{h \rightarrow 0} \frac{\arctan\left(\frac{h}{4 + \sqrt{3}}\right)}{\left(\frac{h}{4 + \sqrt{3}}\right)} \cdot \frac{1}{4 + \sqrt{3}h} + \frac{\pi}{3} \\
 &= \sqrt{3} \cdot 1 \cdot \frac{1}{4} + \frac{\pi}{3} \\
 &= \boxed{\frac{\sqrt{3}}{4} + \frac{\pi}{3}}
 \end{aligned}$$

Problem 3

Suppose f is a one-to-one differentiable function and its inverse function f^{-1} is also differentiable. Use implicit differentiation to show that

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

provided that the denominator is not 0.

$$\begin{aligned}
 &f(f^{-1}(x)) = x \\
 \iff &f'(f^{-1}(x)) \cdot (f^{-1})'(x) = 1 \\
 \iff &\boxed{(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}}
 \end{aligned}$$

Problem 6

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^5 + e^{\frac{x}{2}} + e^{x^3}$ and $g(x) = f^{-1}(x)$. Find $g'(2)$.

$$\begin{aligned}
 2 &= x^5 + e^{\frac{x}{2}} + e^{x^3} \implies g(2) = 0 \\
 g'(2) &= \frac{1}{f'(0)} = \frac{1}{\frac{1}{2}e^0 + 3 \cdot 0^2 \cdot e^{0^3}} = \boxed{2}
 \end{aligned}$$

Problem 7

Find the derivatives of the following inverse trigonometric functions:

(e) $y = \cot^{-1} x$

$$\begin{aligned}\cot y &= x \\ y' &= -\frac{1}{\csc^2 x} \\ y' &= -\frac{1}{x^2 + 1} \\ \cot^{-1} x &= -\frac{1}{x^2 + 1}\end{aligned}$$

(f) $y = \csc^{-1} x$

$$\begin{aligned}\csc y &= x \\ y' &= -\frac{1}{\csc x \cot x} \\ y' &= -\frac{1}{|x|\sqrt{x^2 - 1}} \\ \csc^{-1} x &= -\frac{1}{|x|\sqrt{x^2 - 1}}\end{aligned}$$

Problem 8

Differentiate

(a) $y = x^2 \arcsin x$

$$\frac{dy}{dx} = 2x \arcsin x + \frac{x^2}{\sqrt{1 - x^2}}$$

(b) $y = \frac{1 + \arctan x}{2 - 3 \arctan x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(2 - 3 \arctan x) \frac{1}{x^2 + 1} - (1 + \arctan x) \left(-\frac{3}{x^2 + 1}\right)}{(2 - 3 \arctan x)^2} \\ &= \frac{\frac{1}{x^2 + 1} (2 - 3 \arctan x + 3 + 3 \arctan x)}{(2 - 3 \arctan x)^2} \\ &= \frac{5}{(x^2 + 1)(2 - 3 \arctan x)^2}\end{aligned}$$

(c) $f(x) = \sec^{-1} x \cdot \csc^{-1} x$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\csc^{-1} x}{|x|\sqrt{x^2 - 1}} - \frac{\sec^{-1} x}{|x|\sqrt{x^2 - 1}} \\ &= \frac{\csc^{-1} x - \sec^{-1} x}{|x|\sqrt{x^2 - 1}}\end{aligned}$$

(d) $f(x) = x(\tan^{-1} x)^2$

$$\begin{aligned}\frac{dy}{dx} &= (\arctan x)^2 + x \cdot 2 \arctan x \cdot \frac{1}{x^2 + 1} \\ &= \boxed{\arctan x \left(\arctan x + \frac{2x}{x^2 + 1} \right)}\end{aligned}$$

(e) $f(x) = (\arcsin(x^3))^4$

$$\begin{aligned}\frac{dy}{dx} &= 4(\arcsin(x^3))^3 \cdot \frac{1}{\sqrt{1-x^6}} \cdot 3x^2 \\ &= \boxed{\frac{12x^2(\arcsin x^3)^3}{\sqrt{1-x^6}}}\end{aligned}$$

(f) $y = \arctan(e^{-x^2})$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{e^{-2x^2} + 1} \cdot e^{-x^2} \cdot (-2x) \\ &= \boxed{-\frac{2xe^{-x^2}}{e^{-2x^2} + 1}}\end{aligned}$$

Problem 9

Let $y = \arctan \sqrt{\frac{a-x}{a+x}}$, $-a < x < a$. Find $\frac{dy}{dx}$.

Hint: Use substitution: $x = a \cos \theta$.

$$\begin{aligned}x &= a \cos \theta \implies \theta = \arccos \frac{x}{a} \\ y &= \arctan \sqrt{\frac{a-x}{a+x}} = \arctan \sqrt{\frac{a(1-\cos \theta)}{a(1+\cos \theta)}} = \arctan(\tan \frac{\theta}{2}) = \frac{\theta}{2} \\ \frac{dy}{dx} &= \frac{1}{2} \cdot \frac{d\theta}{dx} = \frac{1}{2} \left(-\frac{1}{\sqrt{1-\frac{x^2}{a^2}}} \right) \left(\frac{1}{a} \right) = \boxed{-\frac{1}{2\sqrt{a^2-x^2}}}\end{aligned}$$

Problem 11

Find an equation of the line tangent to the graph of $y = \ln(x^2 + 4) - x \arctan\left(\frac{x}{2}\right)$ at $x = 2$.

$$\begin{aligned}f(2) &= \ln(8) - 2 \arctan 1 = 3 \ln 2 - \frac{\pi}{2} \\ f'(x) &= \frac{2x}{x^2 + 4} - \arctan\left(\frac{x}{2}\right) - x \cdot \frac{\frac{1}{2}}{1 + \frac{x^2}{4}} \\ f'(2) &= \frac{1}{2} - \frac{\pi}{4} - \frac{1}{2} = -\frac{\pi}{4} \\ &\boxed{y - 3 \ln 2 + \frac{\pi}{2} = -\frac{\pi}{4}(x - 2)}\end{aligned}$$

Problem 12

Evaluate $\lim_{h \rightarrow 0} \frac{(a+h)^2 \arcsin(a+h) - a^2 \arcsin a}{h}$.

$$\begin{aligned}
 & \lim_{h \rightarrow 0} \frac{(a+h)^2 \arcsin(a+h) - a^2 \arcsin a}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(a^2 + 2ah + h^2) \arcsin(a+h) - a^2 \arcsin a}{h} \\
 &= \lim_{h \rightarrow 0} \frac{a^2(\arcsin(a+h) - \arcsin a)}{h} + \lim_{h \rightarrow 0} \frac{2ah \arcsin(a+h)}{h} + \lim_{h \rightarrow 0} \frac{h^2 \arcsin(a+h)}{h} \\
 &= a^2 \cdot \frac{1}{\sqrt{1-a^2}} + \lim_{h \rightarrow 0} 2a \arcsin(a+h) + \lim_{h \rightarrow 0} h \arcsin(a+h) \\
 &= \boxed{\frac{a^2}{\sqrt{1-a^2}} + 2a \arcsin a}
 \end{aligned}$$